

Week Report 3

Completed Work for week 3

- lab 3
- notes 3

Practice Screenshots

Practice 3

1: emp@cis106: ~

2: emp@cis106: ~

emp@cis106:~\$

Every 2.0s: date
Thu Feb 26 05:53:45 PM EST 2026

The Bash Shell - Google S

docs.google.com/presentation/d/e/2PACX-1vSMW_nt1pQzuvuV3HIZ-3gw9...

Practice 3: Basic Shell Usage Commands

1. Use the **echo** command to print a line of text to the screen

echo "Hello Class!"

2. Use the **sleep** command to put the shell to sleep for 3 seconds

sleep 3

3. Use the **watch** command to repeat the date command every 2 seconds (Press CTRL + C to exit)

watch date

4. Use the **history** command to see a history of all the commands you ran so far

history

5. Use the **clear** command to clear the screen

clear

Practice 4

1: emp@cis106: ~

Change delay from 3.0 to

UID	USER	PR	NI	VI	RES	SHR	S
1837	emp	20	0	3831760	267392	139660	S
3590	emp	20	0	582680	76404	56972	S
4563	root	20	0	93332	23824	20172	S
4565	root	20	0	93332	23840	20200	S
1935	emp	20	0	385612	11852	7004	S
2329	emp	20	0	615392	40452	31412	S
27	root	20	0	0	0	0	S
3312	emp	20	0	2456620	101660	79648	S
3935	emp	20	0	2425392	77680	63960	S
4051	root	20	0	0	0	0	I
4553	root	20	0	87288	12824	9500	S
1	root	20	0	24292	14980	10780	S
2	root	20	0	0	0	0	S
3	root	20	0	0	0	0	S
4	root	0	-20	0	0	0	I
5	root	0	-20	0	0	0	I
6	root	0	-20	0	0	0	I
7	root	0	-20	0	0	0	I
8	root	0	-20	0	0	0	I
10	root	20	0	0	0	0	I
11	root	0	-20	0	0	0	I
13	root	0	-20	0	0	0	I
14	root	20	0	0	0	0	I
15	root	20	0	0	0	0	I
16	root	20	0	0	0	0	I
17	root	20	0	0	0	0	S
18	root	20	0	0	0	0	I
19	root	20	0	0	0	0	S
20	root	20	0	0	0	0	S
21	root	rt	0	0	0	0	S
22	root	-51	0	0	0	0	S
23	root	20	0	0	0	0	S
24	root	20	0	0	0	0	S
25	root	-51	0	0	0	0	S
26	root	rt	0	0	0	0	S
29	root	0	-20	0	0	0	I
31	root	20	0	0	0	0	I
32	root	20	0	0	0	0	S
33	root	0	-20	0	0	0	I
34	root	20	0	0	0	0	S

emp@cis106:~\$ ^C
emp@cis106:~\$ du -hd1 /home
1.1G /home/emp
1.1G /home
emp@cis106:~\$

The Bash Shell - Google S

docs.google.com/presentation/d/e/2PACX-1vSMW_nt1pQzuvuV3HIZ-3gw9...

Practice 4: Basic System Information Commands

Let's use the commands in the previews table to answer some questions:

1. What is the username of the currently logged on user? **whoami**

2. How long has the computer been running for? **uptime**

3. What is my computer name (node name/hostname)? **hostname**

4. How much free memory do I have? **free -h**

5. How much disk space do I have available? **df -h /**

6. Can I see all the process and programs running on my computer (CTRL + C to exit)? **top**

7. Can I print general information about my system? **uname -a**

8. Can I see how much space my home folder is occupying? **du -hd1 /home**

Practice 5

1: emp@cis106: ~

Feb 26 5:55 PM

Tilix: emp@cis106: ~

ip_tables 28672 0

x_tables 53248 1 ip_tables

autofs4 57344 2

ext4 1142784 1

crc16 12288 1 ext4

mbcache 16384 1 ext4

jbd2 208704 1 ext4

crc32c_generic 12288 0

hid_generic 12288 0

usbhid 77824 0

hid 262144 2 usbhid, hid_generic

sr_mod 28672 0

cdrom 81920 1 sr_mod

sd_mod 81920 2

ata_generic 12288 0

ahci 49152 2

vmwgfx 471040 7

ata_piix 45056 0

libahci 61440 1 ahci

drm_ttm_helper 16384 2 vmwgfx

ttm 106496 2 vmwgfx, drm_ttm_helper

ohci_pci 20480 0

libata 462848 4 ata_piix, libahci, ahci, ata_generic

ehci_pci 16384 0

drm_kms_helper 253952 2 vmwgfx, drm_ttm_helper

ehci_hcd 118592 1 ehci_pci

ohci_hcd 65536 1 ohci_pci

drm 774144 12 vmwgfx, drm_kms_helper, drm_ttm_helper, ttm

scsi_mod 327680 4 sd_mod, libata, sg, sr_mod

e1000 180224 0

usbcore 409600 5 ohci_hcd, ehci_pci, usbhid, ehci_hcd, ohci_pci

psmouse 217088 0

crc32_pcmul 12288 0

serio_raw 16384 0

crc32c_intel 16384 2

i2c_piix4 28672 0

i2c_smbus 16384 1 i2c_piix4

usb_common 16384 3 ohci_hcd, usbcore, ehci_hcd

video 81920 0

vml 28672 1 video

button 24576 0

scsi_common 16384 5 scsi_mod, sd_mod, libata, sg, sr_mod

emp@cis106:~\$ sensors

No sensors found!

Make sure you loaded all the kernel drivers you need.

Try sensors-detect to find out which these are.

emp@cis106:~\$

docs.google.com/presentation/d/e/2PACX-1vSMW_nt1pQzuuV3HIZ-3gw9...

Practice 5: Basic Hardware Information Commands

Let's use the commands in the previews table to answer some questions:

- What brand and model of CPU do I have? **lscpu**
- Is my USB drive/camera/microphone connected? **lsusb**
- Is my Wifi card connected? **lspci**
- How many partitions does my hard drive have? **lsblk**
- How do I get a quick summary of my hardware? **inxi**
- Is XYZ module (driver) loaded? **lsmod**
- Is my computer overheating? **sensors**

Practice 6

1: emp@cis106: ~

Feb 26 5:56 PM

Tilix: emp@cis106: ~

emp@cis106:~\$ fortune

Anyone who has had a bull by the tail knows five or six more things than someone who hasn't.

-- Mark Twain

emp@cis106:~\$ figlet "John Doe"

John Doe

emp@cis106:~\$ Cowsay "hellow world"

bash: Cowsay: command not found

emp@cis106:~\$ cmatrix

bash: sl: command not found

emp@cis106:~\$ echo "John Smith" | rev

> toilet "Bob" --metal

>

docs.google.com/presentation/d/e/2PACX-1vSMW_nt1pQzuuV3HIZ-3gw9...

Practice 6: Basic "Fun" commands

- Use the fortune command to display a couple of random quotes
fortune
- Use the figlet command to display your name in large ASCII letters
figlet "John Doe"
- Use the cowsay command to display a message with the cow character
cowsay "Hello world"
- Enter the Matrix! (CTRL + C to exit)
cmatrix
- Display the running locomotive
sl
- Reverse your name
echo "John Smith" | rev
- Use toilet to display your name in metallic colors
toilet "Bob" --metal

Practice 7

2 / 4

Feb 26 5:58 PM

Tilix: emp@cis106: ~

1/1

```
56 exit
57 echo "Hello Class!"
58 sleep 3
59 sleep 3
60 watch date
61 echo "hello Class!"
62 sleep 3
63 history
64 clear
65 whoami
66 uptime
67 hostname
68 free -h
69 df -h
70 df -h/
71 top
72 du -hd1 /home
73 clear
74 lscpu
75 lsusb
76 lspci
77 lsblk
78 lnxsi
79 lsmod
80 sensors
81 clear
82 fortune
83 figlet "john Doe"
84 cowsay "hello world"
85 cmatrix
86 sl
87 echo "john Smith | rev
88 toilet "Bob" --metal
89 clear
90 clear
91 date
92 echo "hello world"
93 uname -a
94 history
emp@cis106:~$ !#
emp@cis106:~$ echo "hello world"
hello world
emp@cis106:~$ !!world
echo "hello world"world
hello worldworld
emp@cis106:~$
```

docs.google.com/presentation/d/e/2PACX-1vSMW_nt1pQzuuV3HIZ-3gw6

Practice 7: Using the command history

1. Open a terminal window and type the command: **date**
2. Type the command: **echo "hello world"**
3. Type the command: **uname -a**
4. Type the command: **history**
 - a. What do you see?
5. Using the **!#** shortcut, repeat the **echo** command
6. Use the **!!** to repeat the last command
7. Type: **echo "hello"**
8. Type: **!!world**
 - a. What happened?

Practice 8

Feb 26 6:01 PM

Tilix: emp@cis106: ~

1/1

```
NAME(1) User Commands UNAME(1)
NAME
uname - print system information
SYNOPSIS
uname [OPTION]...
DESCRIPTION
Print certain system information. With no
OPTION, same as -s.
-a, --all
    print all information, in the follow-
    ing order, except omit -p and -i if
    unknown:
-s, --kernel-name
    print the kernel name
-n, --nodename
    print the network node hostname
-r, --kernel-release
    print the kernel release
-v, --kernel-version
    print the kernel version
-m, --machine
    print the machine hardware name
-p, --processor
    print the processor type
    (non-portable)
-i, --hardware-platform
    print the hardware platform
    (non-portable)
-o, --operating-system
    print the operating system
--help display this help and exit
... MOST: *stdin* (1.1)8%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.
```

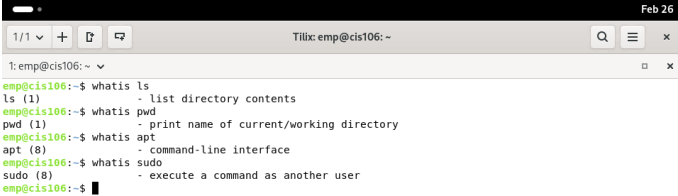
2: emp@cis106: ~

```
emp@cis106:~$ free -g
              total        used        free      sh
red buff/cache  available
Mem:           7          2          4
0              1          5
Swap:          2          0          2
emp@cis106:~$
```

Practice 8: Using man

1. Open 2 terminal windows or split tilix into 2
2. In one terminal type the command: **man uname**
3. On the second terminal, and using the information in the manual page, find out the following information about your system:
 - a. What is the kernel name
 - b. What is the hostname
 - c. What is the hardware platform and operating system
4. Open the man page for each of these command:
 - a. **date**, **df**, **free**, **clear**, **history**
5. Using the man page of the **free** command, display the memory information in gigabytes

Practice 9



```

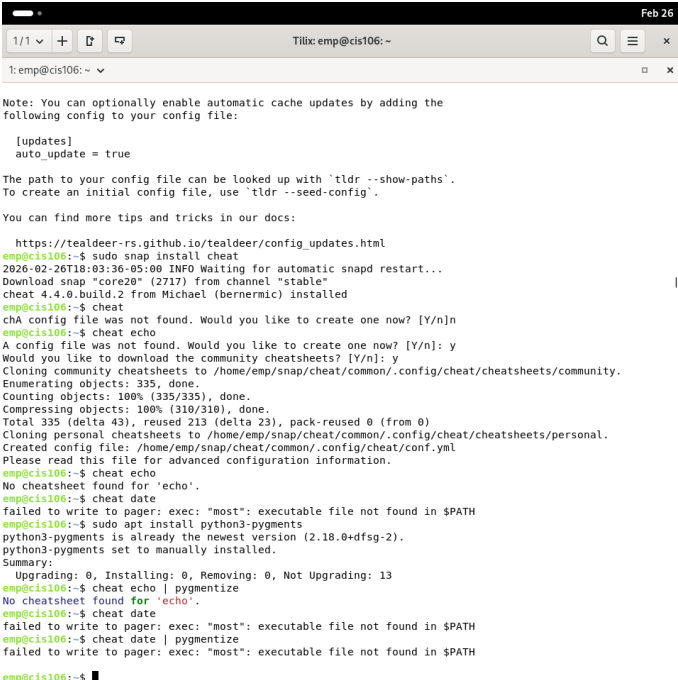
1: emp@cis106: ~
emp@cis106:~$ whatis ls
ls (1)                - list directory contents
emp@cis106:~$ whatis pwd
pwd (1)               - print name of current/working directory
emp@cis106:~$ whatis apt
apt (8)               - command-line interface
emp@cis106:~$ whatis sudo
sudo (8)              - execute a command as another user
emp@cis106:~$

```

Practice 9: Using the help option

1. A lot of times all you need is to remember the syntax or the options of a command. In this scenario, you can simply use the **--help** option. Remember that for some commands you have to use **-h** instead of **--help**
2. Display the help options for the following commands:
 - a. **free**
 - b. **man**
 - c. **date**
3. Use the **whatis** command to find out what these commands do:
 - a. **ls**
 - b. **pwd**
 - c. **apt**
 - d. **sudo**

Practice 10



```

1: emp@cis106: ~
Note: You can optionally enable automatic cache updates by adding the
following config to your config file:

[updates]
auto_update = true

The path to your config file can be looked up with 'tldr --show-paths'.
To create an initial config file, use 'tldr --seed-config'.

You can find more tips and tricks in our docs:
https://tealdeer-rs.github.io/tealdeer/config_updates.html
emp@cis106:~$ sudo snap install cheat
2026-02-26T18:03:36-05:00 INFO Waiting for automatic snapd restart...
Download snap "core20" (2717) from channel "stable"
cheat 4.4.0.build.2 from Michael (bernermic) installed
emp@cis106:~$ cheat
chA config file was not found. Would you like to create one now? [Y/n]n
emp@cis106:~$ cheat echo
A config file was not found. Would you like to create one now? [Y/n]: y
Would you like to download the community cheatsheets? [Y/n]: y
Cloning community cheatsheets to /home/emp/snap/cheat/common/.config/cheat/cheatsheets/community.
Enumerating objects: 335, done.
Counting objects: 100% (335/335), done.
Compressing objects: 100% (310/310), done.
Total 335 (delta 43), reused 213 (delta 23), pack-reused 0 (from 0)
Cloning personal cheatsheets to /home/emp/snap/cheat/common/.config/cheat/cheatsheets/personal.
Created config file: /home/emp/snap/cheat/common/.config/cheat/conf.yml
Please read this file for advanced configuration information.
emp@cis106:~$ cheat echo
No cheatsheet found for 'echo'.
emp@cis106:~$ cheat date
failed to write to pager: exec: "most": executable file not found in $PATH
emp@cis106:~$ sudo apt install python3-pygments
python3-pygments is already the newest version (2.18.0+dfsg-2).
python3-pygments set to manually installed.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 13
emp@cis106:~$ cheat echo | pygmentize
No cheatsheet found for 'echo'.
emp@cis106:~$ cheat date
failed to write to pager: exec: "most": executable file not found in $PATH
emp@cis106:~$ cheat date | pygmentize
failed to write to pager: exec: "most": executable file not found in $PATH
emp@cis106:~$

```

Practice 10: Tldr & Cheat

1. Install the tealdeer application using this command: **sudo apt install tealdeer -y**
2. Then update the cache with: **tldr -u**
3. Display the available examples for the **echo** and **data** commands
4. Install the snap package for cheat using this command: **sudo snap install cheat**
5. Run the **cheat** command to download the available cheats
6. Look at the available cheats for the **git** and **date** command
7. Now let's add some syntax highlighting. Install the package python3-pygments with the command: **sudo apt install python3-pygments -y**
8. Repeat the commands:
 - a. **cheat git | pygmentize**
 - b. **cheat data | pygmentize**